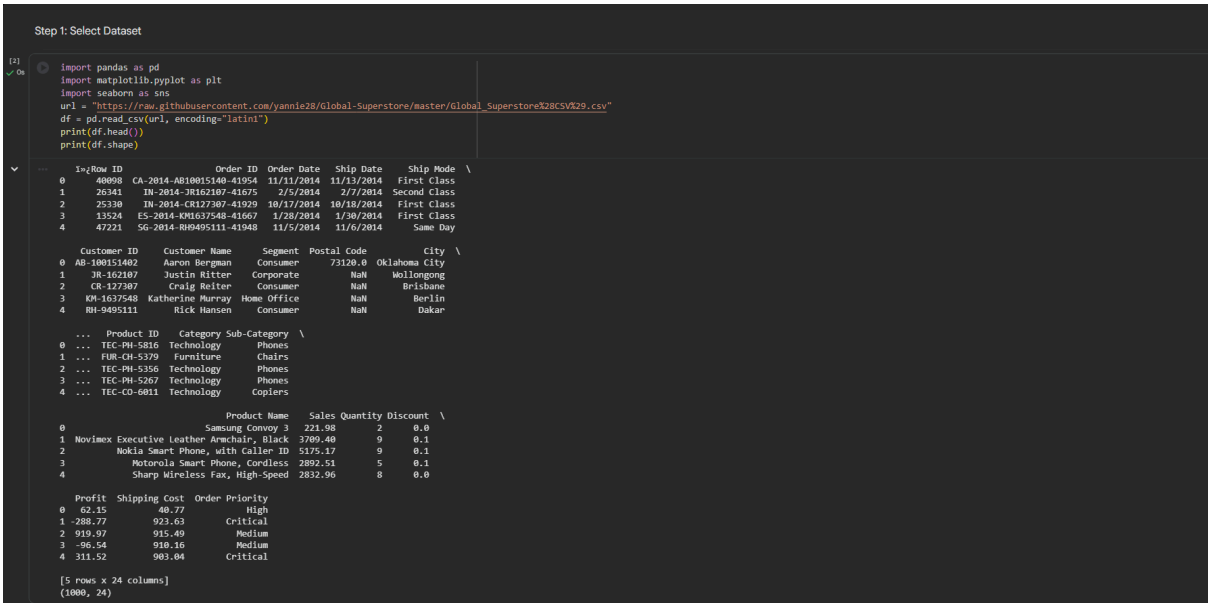
LAB SHEET 9

Experiment Name: Data Storytelling and Business Insight Generation using Interactive Visualizations

Steps :

1. Select a real-world dataset such as the Global Superstore Sales Dataset, Netflix Dataset, World Happiness Report, or COVID-19 Dataset.
2. Analyze the dataset to identify the primary business problem or analytical objective.
3. Perform exploratory data analysis to discover significant trends, patterns, correlations, and anomalies.
4. Develop a set of meaningful visualizations such as bar charts, line graphs, heatmaps, and dashboards to support the analysis.
5. Identify Key Performance Indicators (KPIs) and highlight important business metrics relevant to the selected dataset.
6. Arrange the visualizations in a logical sequence to create a coherent data story that explains the problem, analysis, findings, and recommendations.
7. Interpret the visualizations and identify major insights, opportunities, risks, or performance gaps.
8. Prepare actionable recommendations based on the analytical findings to support business decision-making.
9. Present the complete data story using presentation slides, Tableau, Power BI, or a Jupyter Notebook.
10. Discuss how effective storytelling improves communication between data analysts and business stakeholders.

Solution ->



Step 1: Select Dataset

```
[2]
✓ 0s
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
url = "https://raw.githubusercontent.com/yannie28/Global-Superstore/master/Global_Superstore%28CSV%29.csv"
df = pd.read_csv(url, encoding="latin1")
print(df.head())
print(df.shape)
```

```
   Ï»¿Row ID         Order ID  Order Date   Ship Date       Ship Mode  \
0      40098  CA-2014-AB10015140-41954  11/11/2014  11/13/2014   First Class
1      26341     IN-2014-JR162107-41675    2/5/2014   2/7/2014  Second Class
2      25330     IN-2014-CR127307-41929  10/17/2014  10/18/2014   First Class
3      13524    ES-2014-KM1637548-41667   1/28/2014   1/30/2014   First Class
4      47221    SG-2014-RH9495111-41948   11/5/2014   11/6/2014      Same Day

    Customer ID     Customer Name      Segment  Postal Code          City  \
0  AB-10015140     Aaron Bergman     Consumer      73120.0  Oklahoma City
1     JR-162107     Justin Ritter    Corporate          NaN     Wollongong
2     CR-127307      Craig Reiter     Consumer          NaN       Brisbane
3    KM-1637548  Katherine Murray  Home Office          NaN         Berlin
4    RH-9495111       Rick Hansen     Consumer          NaN          Dakar

   ...   Product ID    Category Sub-Category  \
0  ...  TEC-PH-5816  Technology       Phones
1  ...  FUR-CH-5379   Furniture       Chairs
2  ...  TEC-PH-5356  Technology       Phones
3  ...  TEC-PH-5267  Technology       Phones
4  ...  TEC-CO-6011  Technology      Copiers

                                Product Name    Sales  Quantity  Discount  \
0                    Samsung Convoy 3   221.98         2       0.0
1  Novimex Executive Leather Armchair, Black  3709.40         9       0.1
2         Nokia Smart Phone, with Caller ID  5175.17         9       0.1
3            Motorola Smart Phone, Cordless  2892.51         5       0.1
4            Sharp Wireless Fax, High-Speed  2832.96         8       0.0

   Profit  Shipping Cost  Order Priority
0   62.15          40.77            High
1 -288.77         923.63        Critical
2  919.97         915.49          Medium
3  -96.54         910.16          Medium
4  311.52         903.04        Critical

[5 rows x 24 columns]
(1000, 24)
```

Step 2: Identify Business Problem

```
print("Objective: Analyze sales, profit, regions, categories and trends")
print("to identify business opportunities and performance gaps.")
```

```
Objective: Analyze sales, profit, regions, categories and trends
to identify business opportunities and performance gaps.
```

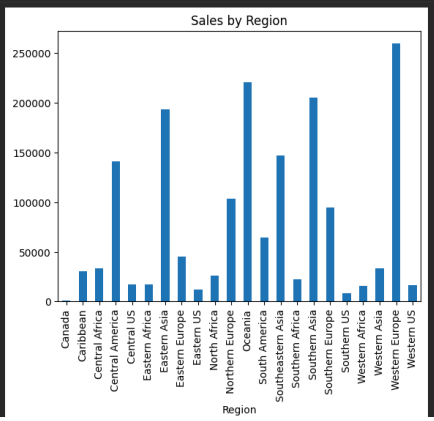
Step 3 — Exploratory Data Analysis

```
print(df.info())
print(df.describe())
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 24 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   1=Row ID        1000 non-null   int64
 1   Order ID        1000 non-null   object
 2   Order Date      1000 non-null   object
 3   Ship Date       1000 non-null   object
 4   Ship Mode       1000 non-null   object
 5   Customer ID     1000 non-null   object
 6   Customer Name   1000 non-null   object
 7   Segment         1000 non-null   object
 8   Postal Code     194 non-null    float64
 9   City            1000 non-null   object
 10  State           1000 non-null   object
 11  Country         1000 non-null   object
 12  Region          1000 non-null   object
 13  Market          1000 non-null   object
 14  Product ID      1000 non-null   object
 15  Category        1000 non-null   object
 16  Sub-Category    1000 non-null   object
 17  Product Name    1000 non-null   object
 18  Sales           1000 non-null   float64
 19  Quantity        1000 non-null   int64
 20  Discount        1000 non-null   float64
 21  Profit          1000 non-null   float64
 22  Shipping Cost   1000 non-null   float64
 23  Order Priority  1000 non-null   object
dtypes: float64(5), int64(2), object(17)
memory usage: 187.6+ KB
None
```

Step 4: Create Visualizations

```
df.groupby("Region")["Sales"].sum().plot(kind="bar", title="Sales by Region")
plt.show()
df.groupby("Category")["Sales"].sum().plot(kind="bar", title="Sales by Category")
plt.show()
df["Order Date"] = pd.to_datetime(df["Order Date"])
df.groupby(df["Order Date"].dt.to_period("M"))["Sales"].sum().plot(
    kind="line", title="Monthly Sales Trend"
)
plt.show()
sns.heatmap(df.select_dtypes("number").corr(), annot=True)
plt.show()
```



Step 5: Identify KPIs

```
print("Total Sales:", df["Sales"].sum())
print("Total Profit:", df["Profit"].sum())
print("Total Quantity:", df["Quantity"].sum())
print("Total Orders:", df["Order ID"].nunique())
print("Total Customers:", df["Customer ID"].nunique())
print("Average Order Value:", df["Sales"].sum() / df["Order ID"].nunique())
```

```
Total Sales: 1710971.47
Total Profit: 288920.44000000006
Total Quantity: 5558
Total Orders: 867
Total Customers: 800
Average Order Value: 1973.4388350634372
```

Step 6: Arrange Data Story

```
print("""
Data Story:
1. Business Problem
2. Sales Overview
3. Regional Performance
4. Category Performance
5. Sales Trend
6. KPIs
7. Key Findings
8. Recommendations
""")
```

```
Data Story:
1. Business Problem
2. Sales Overview
3. Regional Performance
4. Category Performance
5. Sales Trend
6. KPIs
7. Key Findings
8. Recommendations
```

Step 7: Identify Insights

[8] ✓ 0s

```
print("Highest Sales Region:",
      df.groupby("Region")["Sales"].sum().idxmax())
print("Highest Sales Category:",
      df.groupby("Category")["Sales"].sum().idxmax())
print("Highest Profit Category:",
      df.groupby("Category")["Profit"].sum().idxmax())
print("Top Product:",
      df.groupby("Product Name")["Sales"].sum().idxmax())
```

```
Highest Sales Region: Western Europe
Highest Sales Category: Technology
Highest Profit Category: Technology
Top Product: Motorola Smart Phone, Full Size
```

Step 8: Prepare Recommendations

[9] ✓ 0s

```
print("""
Recommendations:
1. Focus marketing on high-performing regions.
2. Promote high-selling products.
3. Review low-profit categories.
4. Optimize inventory using sales trends.
5. Monitor sales and profit KPIs regularly.
""")
```

```
Recommendations:
1. Focus marketing on high-performing regions.
2. Promote high-selling products.
3. Review low-profit categories.
4. Optimize inventory using sales trends.
5. Monitor sales and profit KPIs regularly.
```

Step 9: Present Complete Data Story

[10] ✓ 0s

```
print("""
Presentation Structure:
• Introduction
• Business Problem
• Dataset
• EDA
• Visualizations
• KPIs
• Insights
• Recommendations
• Conclusion
""")
```

```
Presentation Structure:
• Introduction
• Business Problem
• Dataset
• EDA
• Visualizations
• KPIs
• Insights
• Recommendations
• Conclusion
```

Step 10: Importance of Data Storytelling

[11] ✓ 0s

```
print("""
Data storytelling converts analytical results into an understandable
business narrative. It helps stakeholders quickly understand trends,
problems, opportunities and recommended actions.
""")
```

```
Data storytelling converts analytical results into an understandable
business narrative. It helps stakeholders quickly understand trends,
problems, opportunities and recommended actions.
```